

Instructions:

- Make sure to show each step of your solution for full credit - correct answers without work will not receive credit. Some questions may remind you to show your work, but you should assume each question requires you to justify your steps by making your thought process clear in your solution. You can use the solutions we write in class as an indication of how much work you are expected to show. You can ask the instructor if it is not clear how much work you are expected to show on a particular problem.
- Write all of your work and answers on this document. If you need extra paper, please ask the instructor.
- Indicate your affirmation of the following statement by signing below.

I did not give or receive any unauthorized assistance on this exam. I understand that the penalty for academic dishonesty is an F for the course and possible suspension or expulsion from the university.

Signature: _____

Derivatives

$$\begin{aligned}\frac{d}{dx}(\sin^{-1} x) &= \frac{1}{\sqrt{1-x^2}} \\ \frac{d}{dx}(\cos^{-1} x) &= -\frac{1}{\sqrt{1-x^2}} \\ \frac{d}{dx}(\tan^{-1} x) &= \frac{1}{1+x^2} \\ \frac{d}{dx}(\sinh^{-1} x) &= \frac{1}{\sqrt{1+x^2}} \\ \frac{d}{dx}(\cosh^{-1} x) &= \frac{1}{\sqrt{x^2-1}} \\ \frac{d}{dx}(\tanh^{-1} x) &= \frac{1}{1-x^2}\end{aligned}$$

Identities

$$\begin{aligned}\cos^2 x + \sin^2 x &= 1 \\ \tan^2 x + 1 &= \sec^2 x \\ \sin^2 x &= \frac{1}{2}(1 - \cos(2x)) \\ \cos^2 x &= \frac{1}{2}(1 + \cos(2x)) \\ \cosh^2 x - \sinh^2 x &= 1 \\ 1 - \tanh^2 x &= \operatorname{sech}^2 x\end{aligned}$$

Integrals

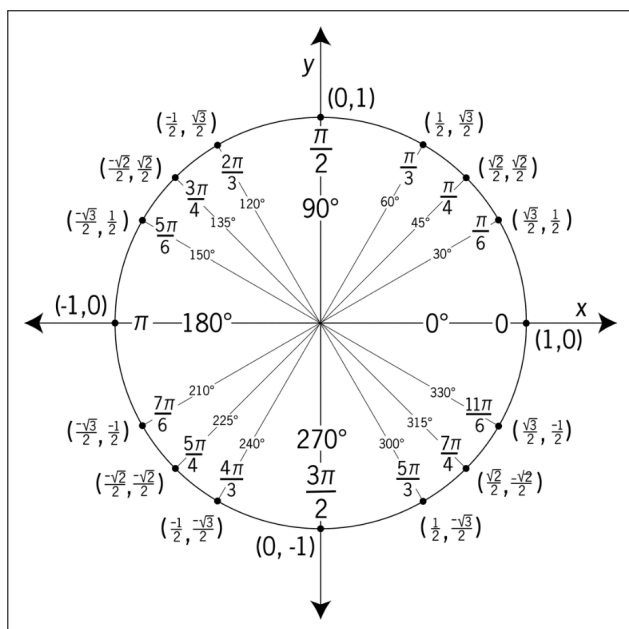
$$\begin{aligned}\int u dv &= uv - \int v du \\ \int \tan x dx &= \ln |\sec x| + C \\ \int \sec x dx &= \ln |\sec x + \tan x| + C \\ \int \frac{dx}{x^2 + a^2} &= \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C\end{aligned}$$

Approx. integration

$$\begin{aligned}|E_T| &\leq \frac{K(b-a)^3}{12n^2} \\ |E_M| &\leq \frac{K(b-a)^3}{24n^2} \\ |E_S| &\leq \frac{L(b-a)^5}{180n^4}\end{aligned}$$

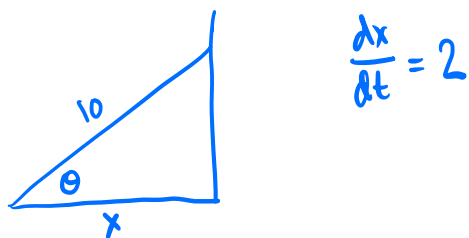
Hyperbolic functions

$$\begin{aligned}\cosh(x) &= \frac{e^x + e^{-x}}{2} \\ \sinh(x) &= \frac{e^x - e^{-x}}{2} \\ \sinh^{-1} x &= \ln(x + \sqrt{x^2 + 1}) \\ \cosh^{-1} x &= \ln(x + \sqrt{x^2 - 1}) \\ \tanh^{-1} x &= \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)\end{aligned}$$



A1 - Inverse trigonometric and hyperbolic functions (inspired by Brightspace problem 4 and part 2 of the class notes)

- (a) Suppose that a ladder 10 feet in length leans against a wall, and the bottom of the ladder begins sliding away from the bottom of the wall at 2 feet per second. What is the rate of change of the angle between the ladder and the floor when the bottom of the ladder is 6 feet from the bottom of the wall? You do not need to simplify your answer, but you should include units.



$$\theta = \cos^{-1}\left(\frac{x}{10}\right)$$

$$\frac{d\theta}{dt} = \frac{-1}{\sqrt{1 - \left(\frac{x}{10}\right)^2}} \cdot \frac{1}{10} \cdot \frac{dx}{dt} \quad @ \ x=6 \quad \rightarrow \quad \frac{-1}{\sqrt{1 - \left(\frac{6}{10}\right)^2}} \cdot \frac{1}{10} \cdot 2 \quad \text{rad/s}$$

- (b) Find the derivative of $g(x) = \sinh^{-1}(\sin(x))$. You don't need to simplify your answer.

$$g'(x) = \frac{1}{\sqrt{1 + \sin^2(x)}} \cdot \cos(x)$$

- (c) Evaluate $\int \frac{\cosh(\sqrt{x})}{2\sqrt{x}} dx$.

$$u = \sqrt{x} \quad du = \frac{1}{2\sqrt{x}} dx \quad \int \cosh(u) du = \sinh(u) + C = \sinh(\sqrt{x}) + C$$

A2 - L'Hospital's rule (inspired by the class notes and Brightspace problem 2)

Evaluate each of the following limits.

(a) $\lim_{x \rightarrow 0} (1 + \sin^2 x)^{\frac{1}{2x^2}} = L$

$$\ln(L) = \lim_{x \rightarrow 0} \frac{\ln(1 + \sin^2 x)}{2x^2} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{2 \sin x \cos x}{4x} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{(1 + \sin^2 x) \cdot (2 \cos^2 x - 2 \sin^2 x) - (2 \sin x \cos x)^2}{(1 + \sin^2 x)^2}$$
$$= \frac{1 \cdot (2 - 0) - 0^2}{4} = \frac{1}{2}$$

$$\rightarrow \boxed{L = e^{\frac{1}{2}}}$$

(b) $\lim_{x \rightarrow 0} \frac{\tan^{-1}(x) - x}{x^3}$

$$\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\frac{1}{1+x^2} - 1}{3x^2} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\frac{-2x}{(1+x^2)^2}}{6x} = \lim_{x \rightarrow 0} \frac{-1}{3(1+x^2)^2} = \boxed{-\frac{1}{3}}$$

A3 - Integration techniques I (inspired by past final worksheet problem 5 and part 2 of the class notes)

Evaluate each of the following integrals.

(a) $\int e^{3\theta} \sin \theta \, d\theta$

$$u = \sin \theta \quad v = \frac{1}{3} e^{3\theta}$$

$$du = \cos \theta \, d\theta \quad dv = e^{3\theta} \, d\theta$$

$$= \frac{1}{3} e^{3\theta} \sin \theta - \frac{1}{3} \int e^{3\theta} \cos \theta \, d\theta$$

$$u = \cos \theta \quad v = \frac{1}{3} e^{3\theta}$$

$$du = -\sin \theta \, d\theta \quad dv = e^{3\theta} \, d\theta$$

$$\int e^{3\theta} \sin \theta \, d\theta = \frac{1}{3} e^{3\theta} \sin \theta - \frac{1}{3} \left(\frac{1}{3} e^{3\theta} \cos \theta + \frac{1}{3} \int e^{3\theta} \sin \theta \, d\theta \right) = \frac{1}{3} e^{3\theta} \sin \theta - \frac{1}{9} e^{3\theta} \cos \theta - \frac{1}{9} \int e^{3\theta} \sin \theta \, d\theta$$

$$\frac{10}{9} \int e^{3\theta} \sin \theta \, d\theta = \frac{1}{3} e^{3\theta} \sin \theta - \frac{1}{9} e^{3\theta} \cos \theta \rightarrow \int e^{3\theta} \sin \theta \, d\theta = \boxed{\frac{3}{10} e^{3\theta} \sin \theta - \frac{1}{10} e^{3\theta} \cos \theta + C}$$

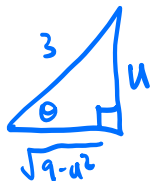
(b) $\int \frac{\sqrt{5-4x-x^2}}{(x+2)^2} \, dx$ (Hint: Complete the square inside of the square root and then use a u -substitution.)

$$= \int \frac{\sqrt{-(x^2+4x-5)}}{(x+2)^2} \, dx = \int \frac{\sqrt{-((x+2)^2-9)}}{(x+2)^2} \, dx = \int \frac{\sqrt{9-(x+2)^2}}{(x+2)^2} \, dx = \int \frac{\sqrt{9-u^2}}{u^2} \, du$$

$$u = 3 \sin \theta$$

$$du = 3 \cos \theta \, d\theta$$

$$\int \frac{\sqrt{9-9\sin^2 \theta}}{9\sin^2 \theta} \cdot 3\cos \theta \, d\theta = \int \cot^2 \theta \, d\theta = \int (\csc^2 \theta - 1) \, d\theta$$



$$= -\cot \theta - \theta + C$$

$$= -\frac{\sqrt{9-u^2}}{u} - \sin^{-1}\left(\frac{u}{3}\right) + C$$

$$= \boxed{-\frac{\sqrt{9-(x+2)^2}}{x+2} - \sin^{-1}\left(\frac{x+2}{3}\right) + C}$$

A4 - Integration techniques II (inspired by past final worksheet problem 2 and Brightspace problem 8)

For each of the following integrals, evaluate it or show that it is divergent.

$$(a) \int \frac{2x^4 + 11x^2 + 2x + 3}{x^3 + 3x} dx = \int \left(2x + \frac{5x^2 + 2x + 3}{x^3 + 3x} \right) dx$$

$$\begin{array}{r} 2x \\ x^3 + 3x \overline{) 2x^4 + 0x^3 + 11x^2 + 2x + 3} \\ \underline{- 2x^4 + 0x^3 + 6x^2} \\ 5x^2 + 2x + 3 \end{array}$$

$$\frac{5x^2 + 2x + 3}{x(x^2 + 3)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 3}$$

$$5x^2 + 2x + 3 = A(x^2 + 3) + (Bx + C)x$$

$$x=0: \quad 3 = A \cdot 3 \rightarrow A=1$$

$$x^2: \quad 5 = A + B \rightarrow B=4$$

$$x: \quad 2 = C$$

$$\int \left(2x + \frac{1}{x} + \frac{4x+2}{x^2+3} \right) dx$$

$$= \int 2x dx + \int \frac{1}{x} dx + \int \frac{4x}{x^2+3} dx + \int \frac{2}{x^2+3} dx$$

$$= x^2 + \ln|x| + 2\ln|x^2+3| + \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{x}{\sqrt{3}}\right) + C$$

$$(b) \int_{-1}^9 (x-4)^{-3} dx$$

$$\lim_{t \rightarrow 4^-} \int_{-1}^t (x-4)^{-3} dx + \lim_{t \rightarrow 4^+} \int_t^9 (x-4)^{-3} dx$$

$$\lim_{t \rightarrow 4^-} \left. -\frac{1}{2} (x-4)^{-2} \right|_{-1}^t + \lim_{t \rightarrow 4^+} \left. -\frac{1}{2} (x-4)^{-2} \right|_t^9$$

$$\lim_{t \rightarrow 4^-} -\frac{1}{2} \left(\underbrace{(t-4)^{-2}}_{\rightarrow \infty} - (-1-4)^{-2} \right) + \lim_{t \rightarrow 4^+} -\frac{1}{2} \left((9-4)^{-2} - \underbrace{(t-4)^{-2}}_{\rightarrow \infty} \right)$$

→ the integral diverges.

A5 - Approximate integration (inspired by Brightspace problem 1 and the class notes)

- (a) Use the Midpoint Rule to approximate $\int_{-1}^1 \sqrt{1+x^3}$ with $n = 4$. You do not need to simplify your answer.

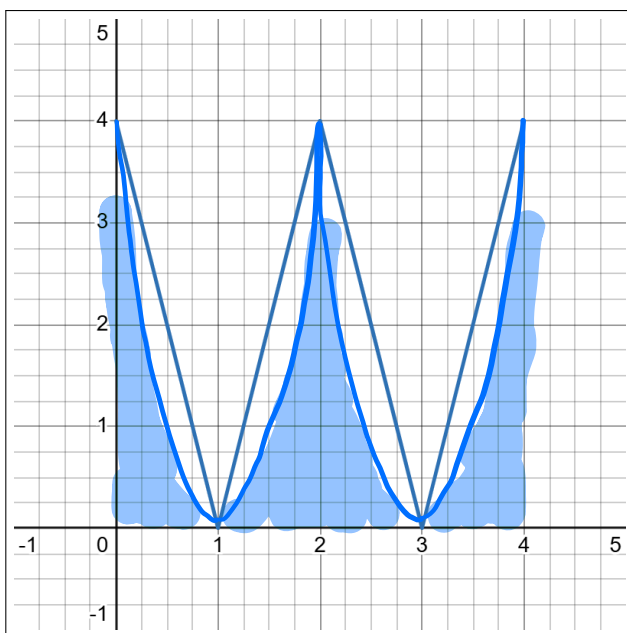
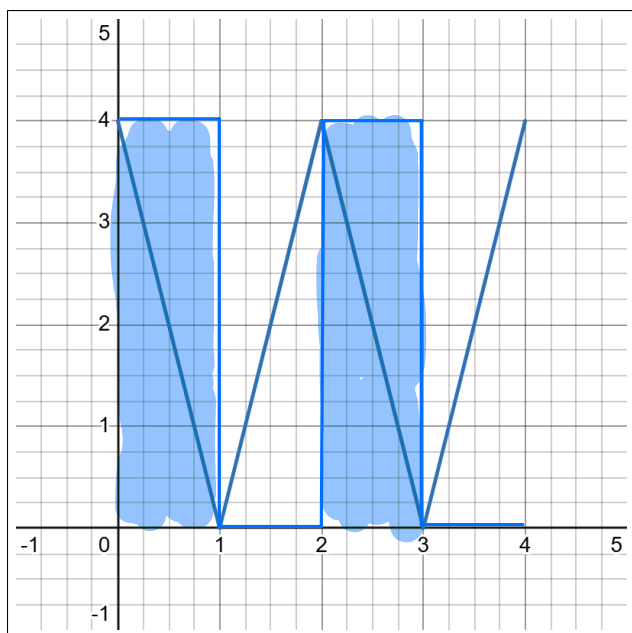


$$0.5 \cdot (\sqrt{1+(-0.75)^3} + \sqrt{1+(-0.25)^3} + \sqrt{1+(0.25)^3} + \sqrt{1+(0.75)^3})$$

- (b) Use Simpson's Rule to approximate $\int_{-1}^1 \sqrt{1+x^3}$ with $n = 4$. You do not need to simplify your answer.

$$\frac{0.5}{3} \left(\sqrt{1+(-1)^3} + 4\sqrt{1+(-0.5)^3} + 2\sqrt{1+0^3} + 4\sqrt{1+(0.5)^3} + \sqrt{1+1^3} \right)$$

- (c) The axes below each show the same function defined on the interval $[0, 4]$.



- (i) On the axes above, shade the area resulting from a left endpoint rule on the interval $[0, 4]$ with $n = 4$. Then state below if this area is an underestimate, an overestimate, or an exact value of the integral.
- (ii) On the axes above, shade in the area resulting from Simpson's rule on the interval $[0, 4]$ with $n = 4$. Then state below if this area is an underestimate, an overestimate, or an exact value of the integral.

Exact value

Underestimate